rather than conducting a continuous layer-wise sweep. Because we purposefully anchored our early layer to the original Liars' Bench baseline and our deeper layer to the maximum theoretical peak before the late-stage performance drop-off identified by Boxo et al. [3], we chose to apply these established bounds directly to our evaluation rather than exhaustively re-proving a continuous layer-wise trajectory from scratch. Second, because our core claims are intentionally conservative—highlighting contextual dependencies ("it depends") rather than declaring absolute algorithmic superiority—we rely on directional shifts in AUROC and refrain from exhaustive statistical significance testing. Third, the lie typology experiments (Figure 2) were restricted to the logistic regression baseline to cleanly isolate distribution effects, and have not yet been evaluated across the more expressive probe families. Fourth, the SAE results were evaluated solely at 66th percentile layer. Fifth, the inherent distribution gap between our synthetic training data (DolusChat) and the complex evaluation prompts (Liars' Bench) likely contributes to the striking anti-transfer observed in the HP datasets. Finally, our discrete classification metrics rely on a default 0.5 probability threshold, whereas future operational work must explore strict False Positive Rate (FPR) constraints.

Acknowledgments

This research was supported by the state of North Rhine-Westphalia as part of the Lamarr Institute for Machine Learning and Artificial Intelligence.

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